



C23-M-401

23445

BOARD DIPLOMA EXAMINATION, (C-23)
MARCH/APRIL—2025
DME – FOURTH SEMESTER EXAMINATION
DESIGN OF MACHINE ELEMENTS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. What are the steps involved in design procedure?
2. A solid shaft is required to transmit a torque of 20 kN-m. Find the necessary diameter of the shaft, if the allowable shear stress is 70 N/mm².
3. Sketch gib-head key with proportionate dimensions.
4. Two parallel shafts 6 meters apart are provided with pulleys of diameters 600 mm and 800 mm respectively. Find the length of belt for turning the two shafts in opposite direction.
5. Write any three differences between belt and chain drive.
6. A wheel has 48 teeth and pitch circle diameter of 384 mm. Find (a) circular pitch and (b) module.
7. Write the different types of gear trains.
8. How do you classify the cams?
9. Define the terms (a) coefficient of fluctuation of energy and (b) coefficient of fluctuation of speed.
10. List the methods of controlling the vibrations in mechanical systems.

PART—B

10×5=50

- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** The cylinder head of a steam engine is subjected to a steam pressure of 0.7 N/mm^2 . It is held in position by means of 12 bolts. A soft copper gasket is used whose $K = 0.5$ to make the joint leak proof. The effective diameter of the cylinder is 300 mm. Considering initial tension due to tightening of the bolt. Find the size of the bolts so that the ultimate stress in the bolts is 800 N/mm^2 and factor of safety as 8.
- 12.** A solid shaft is subjected to a bending moment of 3460 kN-mm and a torque of 1150 kN-mm. Determine the diameter of the shaft given the factor of safety as 6 and assuming ultimate bending stress as 700 N/mm^2 and ultimate shear stress as 520 N/mm^2 .
- 13.** Design a CI flange coupling to connect two shafts in order to transmit 7.5 kW at 720 r.p.m. The following permissible stresses may be assumed :
Permissible shear stress for shaft, bolt and key material = 33 N/mm^2 ,
Permissible crushing stress for bolt and key material = 60 N/mm^2 ,
Permissible shear stress for CI flange = 15 N/mm^2 .
- 14.** A belt transmitting power from a motor to machine weighs 24 N/m and the maximum permissible tension in it is 1000 N. The angle of contact is spread over $5/9$ of the pulley circumference. Coefficient of friction is 0.28. If the belt runs under maximum power condition, determine the optimum belt speed and maximum power transmitted.
- 15.** A three-speed reduction gearbox is to have the following spindle speed ratio as nearly as possible :
Bottom (first) gear $5.2 : 1$; second gear $3 : 1$; third gear $1.6 : 1$; output shaft and input shaft are to be in line and the centre distance which is horizontal between them and the lay shaft is 180 mm. All the gears are to be of 5 mm module and the number of teeth on pinion is 20. Determine the suitable number of teeth on all the gear wheels.

- 16.** Draw the displacement diagram and cam profile to give the following motion to a knife edge follower : Outward stroke through 40 mm during 120° of cam rotation; Dwell for 60° of cam rotation; return stroke during the next 90° . Dwell for the remaining part of cam rotation. The minimum radius of cam is 35 mm. The line of stroke of the follower is coinciding with the centre of the cam axis and the follower moves with SHM. The cam is rotating in clockwise direction.
- 17.** A full journal bearing of 60 mm diameter with length to the diameter ratio as 2 has a bearing pressure of 1.4 MPa. The journal rotates at 800 r.p.m. and the ratio of journal diameter to the diametric clearance is 1000. The lubricating oil working at 75°C has absolute viscosity is 0.011 kg/m-s. End leakage factor is 0.002. The room temperature is 30°C . Find the (a) coefficient of friction, (b) heat generated and (c) heat dissipated, if the coefficient of heat dissipation is $220\text{ W/m}^2/^\circ\text{C}$. Take specific heat of oil is $1850\text{ J/kg-}^\circ\text{C}$.
- 18.** Explain the Porter Governor with a neat sketch.

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